

Lead Data Scientist Finalist Case Study

SePRO Corporation | Water-Resource & Plant-Protection Science

Why we're asking you to do this

You're a finalist for the Lead Data Scientist role at SePRO. This will be SePRO's first data science hire, so it's a foundational role: the person who builds the analytics capability, sets the standards, and earns trust across R&D, Commercial, and Operations. This case study is the capstone of our process and your chance to show how you'd do exactly that.

We'd rather see how you think and build than ask you to grind through several exercises, so this is the one prepared deliverable in our process. You'll present it in person to the team you'd be working with. What we hope to see is your judgment, your rigor, your scientific credibility, and your instinct as a builder.

Time and scope: Please plan for a focused effort, in the range of 6 to 8 hours across the week we've given you. We've kept this to a single deliverable on purpose. A crisp design and a thin working slice will tell us far more than an exhaustive build, so once you've shown your thinking, please stop. Feel free to use whatever public data, libraries, and AI or coding tools you'd normally reach for.

Presentation: You'll present in person to a cross-functional panel that spans scientific, platform, commercial, and operations stakeholders. Plan for roughly 45 minutes to present, followed by about 30 minutes of discussion and Q&A.

About the role, briefly

The Lead Data Scientist builds SePRO's data science capability across three areas: strengthening how research and field-trial data connect to real-world efficacy; building diagnostic and prescriptive tools that turn data into recommendations; and bringing predictive analytics to commercial and operational decisions. Across all three, the throughline is the same: turn complex, fragmented data into decision-ready insight that holds up to scrutiny, and put it into tools people actually use.

This case study centers on the diagnostic and prescriptive part of that mandate, where SePRO's science, data, and customer impact meet most directly.

The challenge

Design an approach to a water-health diagnostic and treatment recommendation that fuses remote-sensing or satellite signals with in-situ aquatic data, is explainable and defensible enough to support a product claim, and could realistically be built into a tool SePRO and its customers would use. Show your

thinking with real public data, build a small working version, and sketch how you would stand it up and scale it.

The most natural problem to anchor on is the harmful algal bloom (HAB): cyanobacteria blooms that threaten drinking-water sources, recreational waters, and aquatic ecosystems, and that SePRO's customers need to get ahead of. Public satellite programs such as EPA, NASA, and NOAA's CyAN detect bloom signals from space, and public in-situ networks report the water chemistry underneath. Fusing the two is exactly the kind of spectral-signal-meets-aquatic-science diagnostic this role is built around, and the operational truth is that timing and targeting matter most: a recommendation made two weeks early, in the right water body, is worth far more than a perfect one made after a bloom is visible.

You have latitude in how you frame the specific question. A few directions that work well:

- **Risk scoring or forecasting:** which water bodies are most likely to bloom in the coming weeks, and what drives that risk?
- **Early-warning detection:** flagging when a water body's combined satellite and in-situ readings depart from its normal pattern.
- **Driver and treatment analysis:** which conditions best explain bloom activity, and what that implies for where, when, and how to intervene.

If you'd rather anchor on an adjacent SePRO-relevant water or plant problem, you're welcome to. We'd just ask that you share your reasoning, still follow the structure below, and keep the spectral-plus-aquatic, claims-defensible spirit of the prompt.

What to deliver

Part A: The analysis

Take the problem through the full data science arc, with your reasoning visible along the way. Our team thinks about this in roughly these stages: define, acquire, prepare, explore, model, evaluate, communicate, and deploy. You don't need to belabor each, but we'd love to see the judgment behind them.

1. **Define it.** The specific question you chose and the decision it informs. Who uses the answer, and what do they do differently because of it?
2. **Acquire and integrate.** Bring together at least two distinct public sources, ideally pairing a remote-sensing or satellite signal with in-situ aquatic or weather data. Tell us where it came from and how you accessed it.
3. **Inspect and prepare.** Show us the data's rough edges, such as gaps, biases, sampling artifacts, and detection limits, and how you handled them. Thoughtful data handling matters to us as much as a clean result.
4. **Explore.** The patterns and relationships you found, and what they ruled in or out. A little care around correlation versus causation goes a long way.

5. **Model.** A method suited to the question. Fit and rigor count for more than sophistication, and a well-validated simple model is very welcome. Be clear about your assumptions, how you validated, and where data leakage or overfitting could creep in.
6. **Evaluate.** How you know the result is meaningful, the metrics that fit the decision, and how confident you are.
7. **Land the “so what.”** Translate it into a recommendation a SePRO user or customer could act on, with a clear-eyed view of where it applies and where it reaches its limits.

A word on defensibility: because SePRO's recommendations can sit behind product claims, we care a lot about whether your approach is explainable and auditable. Wherever you can, show why the model says what it says, not just what it predicts.

Part B: The tool

Build a small, working, interactive tool that puts your analysis in the hands of a non-technical user. Picture a SePRO field or technical specialist, or a customer, who needs the answer rather than the code.

- It should run and be ready to demo live. Running locally or sharing a link are both fine.
- It should draw directly on your Part A analysis, so the two tell one story. A user might select a water body or set of conditions and get back a risk readout, a recommended action window, or the key drivers.
- Please use the stack and tools you'd naturally reach for, whether that's Streamlit, Gradio, a notebook-as-app, a plain web page, a React app, or anything else that lets you move quickly. AI-assisted development is very welcome. We're looking for builder instinct, not a finished product. (Please note: **SePRO will reimburse you up to \$150 of token usage** for the tool build should you wish to vibe code with your preferred LLM.)

Part C: From prototype to platform

Finally, step back and tell us how you'd make this real. A short, practical view (a few slides or a sketch) on:

- **Data architecture:** how you'd connect and scale the inputs, given that SePRO's data lives across research, field-trial, lab, sensor, satellite, distributor, and commercial systems.
- **Production and monitoring:** how you'd deploy it, keep it explainable and auditable, and watch for model drift over time.
- **The first 6 to 12 months:** where you'd start if you were standing up this capability at SePRO, and what you'd build first.

How to present

Pitch your story to a mixed audience: the panel spans scientific, platform, commercial, and operations stakeholders, so part of what we're watching is how you translate between the domain, the data work, and the business. Plan for roughly 45 minutes to present, then about 30

minutes of discussion and Q&A. With your consent, we'll use AI tools to record the session so the full panel can refer back later and focus on your presentation instead of note-taking.

A structure we'd suggest, which you're welcome to adapt:

1. **Problem and decision (about 5 min):** what you chose to explore and why it matters to SePRO.
2. **Data and approach (about 10 min):** your sources, what proved difficult, and your key preparation and modeling choices.
3. **Findings, validation, and defensibility (about 10 min):** what you learned, how confident you are, and why it would hold up behind a claim.
4. **Live tool demo (about 8 min):** show it working, and talk us through your build choices.
5. **From prototype to platform (about 8 min):** architecture, production, monitoring, and your first 6 to 12 months.
6. **Recommendation and discussion (remaining time, then Q&A):** the action you'd suggest, and a conversation with the panel.

Please touch on all six areas, and feel free to shift the time around as your story calls for.

The technical deep dive

After your presentation, we'll set up a shorter, recorded follow-up: a relaxed, screen-sharing walkthrough session where you take us through your code, your data sources, and the analytical choices behind your results. Think of it as a conversation about how you built things rather than a test. With your consent, we'll record the session so the full panel can review your technical depth, since not everyone can join live.

A few ground rules

- **Public data only.** Please use only public data, and steer clear of anything proprietary, confidential, or personal, or anything you don't have the right to use. Using LLM's to curate data is also fine, and you are welcome to synthesize some data if it helps backstop your work in some meaningful way that couldn't be achieved via public real data sources. A citation for each source is all we need.
- **AI and coding tools are welcome.** Use them the way you normally would, and feel free to tell us what you used and where your own judgment shaped the result. We're hiring an analytic builder, and using good tools well is part of that.
- **Reasonable assumptions are fine.** Where SePRO-specific detail would help and you don't have it, make a sensible assumption and tell us what you assumed.
- **Tell us about your trade-offs.** If you took shortcuts or set things aside for time, we'd like to hear about them. Noting a known limitation reflects well on you.

- **Please send materials ahead** of your presentation: your slides (or equivalent) and a short list of the data sources you used, with links, for our reference. You'll demo the work yourself during the session.

Suggested public data sources

These are entirely optional, offered simply so you don't spend your budget hunting for data:

- **EPA Cyanobacteria Assessment Network (CyAN):** satellite-derived cyanobacteria detection for U.S. lakes (the spectral signal), including weekly cyanoHAB forecasting research and links to the underlying Sentinel-3 and Landsat imagery. <https://www.epa.gov/water-research/cyanobacteria-assessment-network-cyan>
- **Water Quality Portal:** in-situ water chemistry from USGS, EPA, and more than 400 agencies, covering nutrients, chlorophyll-a, temperature, and more. <https://www.waterqualitydata.us/>
- **USGS National Water Information System (NWIS):** streamflow, gage, and water data, with dataRetrieval packages available for R and Python. <https://waterdata.usgs.gov/>
- **EPA National Aquatic Resource Surveys / National Lakes Assessment:** standardized lake condition and nutrient data. <https://www.epa.gov/national-aquatic-resource-surveys>
- **NOAA / National Weather Service:** temperature, precipitation, and climate data to pair with water data. <https://www.ncei.noaa.gov/>
- **State environmental and DEQ lake-monitoring programs:** many publish open bloom advisories and monitoring data.

What we're hoping to see

We'll be paying attention to how sharply you framed the problem and tied it to a real decision, the judgment in how you'd source and architect the data, the rigor and honesty of your modeling and validation, whether your approach is explainable and defensible enough to stand behind a claim, the builder instinct behind your tool and your plan to operationalize it, and how clearly you carry a mixed-audience panel and defend your choices. Because this is SePRO's first data science hire, we weight the ability to build production-grade tools and to engage credibly with the science especially highly.

There's no single right answer here. We're far more interested in how you reason and build your way through an open problem than in a polished result.

A few questions you might have

Do I need a background in aquatics or HABs? Not at all. Part of the role is reasoning about an unfamiliar domain and earning scientific credibility quickly, so that's part of what we hope to see. A little reading tends to go a long way.

Can I use a model, library, or AI tool I haven't tried before? Absolutely. Learning quickly and using tools well is a positive sign for us.

What if my analysis doesn't turn up a strong result? That's common and completely fine. A clear account of what the data does and doesn't support is a great outcome, and it's exactly the kind of honesty that holds up behind a claim.

How polished should the tool be? Function comes well before beauty. A thin proof-of-concept that does something useful is exactly what we're hoping for.

How much should I invest in Part C? Keep it short and practical. We want your instincts on architecture, production, and sequencing, not a full architecture document.

Will I need to send you my code or data? No. You keep full ownership of everything you create. You'll walk us through your code and methods live in a recorded screen-share after your presentation, and all we ask in advance is a list of the data sources you used.

How should I split my time? Lean toward the analysis and the thinking. The tool and the prototype-to-platform view should each be a focused, lightweight slice that supports the story.

If anything is unclear before your session, just reply to this email and we'll be glad to help. We're looking forward to seeing how you think.

Warm regards,

The SePRO Hiring Team